

REQUIREMENT ANALYSIS PHASE

Leave Approval System Using ServiceNow

1. Introduction

Requirement Analysis is the process of identifying, analyzing, and documenting the functional and non-functional requirements of the proposed system. This phase helps in understanding the needs of users and defining the system functionalities before development begins.

The Leave Approval System is designed to automate the leave management process within an organization. The system enables employees to submit leave requests and managers to approve or reject them through an automated workflow.

2. Problem Identification

In many organizations, leave requests are handled manually through paper forms, emails, or spreadsheets. These methods create several challenges such as:

- Delays in approval processing.
- Difficulty in tracking leave requests.
- Lack of transparency in approval status.
- Increased administrative workload.
- Risk of data loss and human errors.
- Poor communication between employees and managers.

To overcome these challenges, an automated Leave Approval System is required.

3. Problem Statement

The existing manual leave management process is inefficient, time-consuming, and difficult to track. Employees often experience delays in receiving approval decisions, while managers face challenges in managing multiple leave requests. Therefore, an automated system is required to streamline leave request submission, approval processing, status tracking, and notification management.

4. Stakeholder Analysis

Employee

- Submits leave requests.
- Provides leave details.

- Tracks request status.
- Receives notifications.

Manager

- Reviews leave requests.
- Approves or rejects requests.
- Monitors employee leave records.

Administrator

- Configures and maintains the system.
- Manages workflows and user access.
- Ensures smooth operation of the application.

Organization

- Benefits from improved efficiency.
 - Maintains centralized leave records.
 - Reduces manual administrative effort.
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5. Requirement Gathering

Requirements were gathered by analyzing the leave approval process commonly followed in organizations.

User Requirements

- Employees should be able to submit leave requests.
 - Employees should select leave types.
 - Employees should provide leave dates and reason.
 - Managers should receive approval requests.
 - Managers should approve or reject leave requests.
 - Employees should receive notifications regarding approval decisions.
 - The system should maintain leave records.
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6. Functional Requirements

FR1: Leave Request Submission

The system shall allow employees to submit leave requests.

FR2: Leave Details Entry

The system shall allow employees to enter:

- **Leave Type**
- **Start Date**
- **End Date**
- **Reason**

FR3: Leave Date Validation

The system shall validate leave dates before submission.

FR4: Manager Approval

The system shall send approval requests to the manager.

FR5: Approval Decision

The manager shall be able to approve or reject leave requests.

FR6: Status Management

The system shall update leave request status automatically.

FR7: Notification Management

The system shall generate notifications after approval or rejection.

FR8: Record Management

The system shall store all leave requests in a centralized database.

7. Non-Functional Requirements

Security

Only authorized users should access the system.

Reliability

The system should process leave requests accurately.

Performance

The system should respond quickly to user requests.

Availability

The application should be available whenever users need access.

Scalability

The system should support an increasing number of users and leave requests.

Usability

The interface should be simple and user-friendly.

8. System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet Connection

Software Requirements

- ServiceNow Developer Instance
 - Web Browser (Chrome, Edge, Firefox)
 - Operating System (Windows/Linux/macOS)
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9. Feasibility Analysis

Technical Feasibility

The project is technically feasible because ServiceNow provides built-in tools such as Flow Designer, Tables, Forms, Client Scripts, and Notifications.

Operational Feasibility

The system can be easily used by employees and managers with minimal training.

Economic Feasibility

The project uses a ServiceNow Developer Instance, reducing implementation costs.

Schedule Feasibility

The project can be completed within the planned development schedule.

10. Use Case Analysis

Use Case 1: Submit Leave Request

Actor: Employee

Description: Employee submits a leave request by entering leave details.

Outcome: Leave request is created with Pending status.

Use Case 2: Approve Leave Request

Actor: Manager

Description: Manager reviews the leave request and approves it.

Outcome: Leave request status changes to Approved.

Use Case 3: Reject Leave Request

Actor: Manager

Description: Manager reviews the leave request and rejects it.

Outcome: Leave request status changes to Rejected.

11. Requirement Summary

Requirement ID	Requirement Description
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FR1	Submit Leave Request
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FR2	Enter Leave Details
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FR3	Validate Leave Dates
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FR4	Send Approval Request
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FR5	Approve or Reject Request
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FR6	Update Request Status
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FR7	Generate Notifications
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FR8	Maintain Leave Records
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